File Directory Structure in Linux

A Comprehensive Guide to the Filesystem Hierarchy Standard (FHS)

1. Fundamental Principles

In Linux, a core philosophy governs the entire operating system design:

- **"Everything is a file"** in Linux. If something is not a file, it is a running process.

Categories of Files in Linux

> **1. General Files (Regular Files)**
> Includes documents, images, audio, video files, text files, and binary executables.

> **2. Directory Files**
> Special files (folders) that provide storage space and logical organization for other files and directories.

> **3. Device Files**
> Special files that enable interaction and usability of hardware components on the system. These are located in the `/dev` directory.

2. Linux Directory Structure & FHS

Linux follows a single **hierarchical tree structure** defined by the **Filesystem Hierarchy Standard (FHS)**.

`/` (Root Directory)

- The top-most directory in the Linux File System (LFS).
- All other directories and subdirectories branch out from `/`.
- Every top-level directory under `/` serves a specific, predefined role.

```
/ (Root) ├── /bin ├── /boot ├── /dev ├── /etc ├── /home ├── /lib ├── /media
├── /mnt ├── /opt ├── /sbin ├── /proc ├── /tmp ├── /var ├── /usr ├── /sys
└── /root
```

3. Directory Breakdown & Purpose

Directory	Description & Purpose
`/bin`	Contains essential basic user command binaries (e.g., `ls`, `cp`, `bash`) available for all users.
`/sbin`	Contains system binaries intended for system administration tasks (e.g., `fdisk`, `iptables`).
`/etc`	Contains system-wide configuration files and administrative scripts.
`/usr`	The largest directory; stores installed user applications and software packages. • `/usr/bin` - User programs • `/usr/sbin` - System/administration programs • `/usr/lib` - Libraries for binaries in `/usr/bin` & `/usr/sbin` • `/usr/share` - Shared documentation, architecture-independent data, and icons
`/var`	Stores variable data files whose contents change frequently during operation (e.g., system logs in `/var/log`, caches, databases).
`/home`	Personal home directories for normal users (e.g., `/home/rahul`).
`/root`	Home directory for the root user (the system administrator).
`/tmp`	Stores temporary files created by the system and users. Contents are usually cleared on reboot.
`/boot`	Stores essential files required for the system to boot, such as the Linux kernel (vmlinuz) and GRUB boot loader config files.
`/dev`	Contains device files representing hardware components (e.g., hard drives, USB drives).

Directory	Description & Purpose
`/proc`	Virtual filesystem providing runtime process and kernel parameters (real-time system data).
`/sys`	Virtual filesystem containing system hardware information used by the kernel to manage attached devices.
`/opt`	Reserved for optional and third-party add-on software packages.
`/mnt`	Temporary mount point directory where system administrators can manually mount filesystems.
`/media`	Automatic mount point for removable media devices (e.g., USB drives, CDs, SD cards).

References & Further Reading

- https://medium.com/@santhoshitadakala/linux-directory-structure-and-important-files-paths-explained-e638f407e8b4
- https://www.geeksforgeeks.org/linux-unix/linux-directory-structure/